**Name of a Candidate: _____**

# ST. MARY'S SECONDARY SCHOOL

**2018 MALAWI SCHOOL CERTIFICATE OF EDUCATION MOCK EXAMINATIONS**

# PHYSICAL SCIENCE

**Subject Number: M 162/I**

**Time allowed: 2 hours**

**8:00 – 10:00 am**

**Date:**

## PAPER I

**(100 marks)**

**Instructions:**

1. **This paper contains 10 pages. Please check.**
2. Before beginning, write your **Name** at the top of each page of the question paper
3. Answer all **eight** questions in the spaces provided.
4. The maximum number of marks for each answer is indicated against each question.
5. In the table provided on this page, **tick** against the question number you have answered.

| Question number | Tick if answered | Do not write in these columns | |
|---|---|---|---|
| 1 | | | |
| 2 | | | |
| 3 | | | |
| 4 | | | |
| 5 | | | |
| 6 | | | |
| 7 | | | |
| 8 | | | |
| | | | |

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1. a. Define "kilowatt hour"

_____

_____(1mark)

b. Figure 1 shows a circuit diagram. Study the circuit and use it to answer the questions that follow.

i. If ammeter A reads 0.8 A, calculate the voltage V

(3marks)

ii. If the circuit is switched on for 2 minutes, calculate they themal (heat) energy released by a 5ohm resistor.

(3marks)

**Name of a Candidate: _____**

c. (i) Define 'doping'

_____(1mark)

(ii) State any two differences between intrinsic seminconductors and extrinsic semiconductors.

_____

_____

_____(2marks)

d. (i) a step up transformer has an input supply of 20V and delivers an output of 250V. If the transformer is 80% efficient, work out its secondary current in the secondary circuit, if its primary current is 6.25A.

(3marks)

2. a. Mention any one characteristic of waves

_____(2marks)

b. Figure 1 is a diagram of a wave

i. Name the part shown between P and Q.

_____(1mark)

ii. Calculate the frequency of the wave.

(2marks)

**Name of a Candidate: _____**

iii. Calculate the velocity of the wave.

(2marks)

c. Figure 2 is a diagram showing a mass hanging on a cantilever. If the mass is pulled to point V and released, it vibrates between points V and X through the rest position W.

i. At what point is the kinetic energy of the mass greatest?

_____(1mark)

ii. Why will the vibrations stop after some time?

_____

_____(2marks)

iii. State any two factors that affect the frequency of a cantilever.

_____

_____(2marks)

3. a. Define terminal velocity

_____

_____(2marks)

b. Mention three factors that affect the terminal velocity of an object falling through a liquid.

_____

_____

_____(3marks)

**Name of a Candidate: _____**

c. A car accelerates from 25 $ms^{-1}$ to 75$ms^{-1}$ for 40 seconds. Then it decelerates to 50$ms^{-1}$ in 20 seconds and maintained this speed for another 20 seconds.

i. Sketch the velocity-time graph for this motion.

(3marks)

ii. Calculate the distance covered in the last 40 seconds.

(3marks)

d. Figure 4 shows a resultant of two forces

i. Define resultant force

_____

_____(1mark)

**Name of a Candidate: _____**

ii. Calculate the magnitude of the horizontal and vertical components

| Horizontal component | Vertical component |
|---|---|
| | |

(4marks)

4. a. State any three properties of liquid pressure.

_____

_____

_____(3marks)

b. A small plastic tank of volume $4500cm^3$ contains oil of density $0.78g/cm^3$. If base area of the tank is $50cm^2$, calculate the pressure of oil at the bottom of the tank.

(4marks)

c. In terms of the Kinetic theory of vmatter, explain why wet clothes dry during sunny day.

_____

_____

_____

_____

_____(4marks)

5. (a) Atom **X** is written in short-hand way as $^{39}_{19}\mathbf{X}$.

i. How many neutrons and electrons are there in atom **X** if it is electrically neutral?

_____

_____

(2 marks)

**Name of a Candidate: _____**

ii. How many electrons will be there in the outmost energy level of atom **X**?

_____

(1 mark)

iii. To which period in the Periodic Table does atom **X** belong?

_____

(1 mark)

(b) **Figure 5** below shows nuclear charges of atoms **E** and **F**.

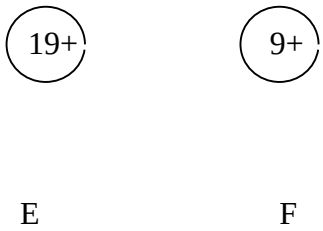


**Figure 5**

i. Complete the diagrams by drawing the energy levels and indicating the number of electrons in each energy level. (4 marks)

ii. What type of bonding takes place when **E** and **F** combine?

_____ (1 mark)

iii. Write the chemical formula of the compound formed when **E** and **F** combine.

_____

(2 marks)

(c) i. Mention **two** physical properties of sulphur.

_____

_____

(1 mark)

ii Name the process used in production of sulphuric acid.

_____

(1 mark)

6. (a) i. What does the symbol $\rightleftharpoons$ mean in a chemical equation?

_____

(1 mark)

ii. Draw a well labelled energy level diagram for the reaction

$2H_2O_{(l)} \longrightarrow 2H_{2(g)} + O_{2(g)}$. $\triangle H = +575kJ/mol$

**Name of a Candidate: _____**

(3 marks)

(b). Given that a chemical reaction has the following:

1. Reducing half equation of $2Ag^{+}(aq) + 2e^{-} \rightarrow 2Ag^{\circ}(s)$
2. Oxidizing half equation of $Mg^{\circ}(s) - 2e^{-} \rightarrow Mg^{2+}(aq)$

i. Write a full ionic equation of the half equations above.

_____

(1mark)

ii. Write the reducing agent and the reduced compound in the equation written (i) above.

_____

_____

(2 marks)

(c). Calculate an empirical formula of organic compound containing 0.12g of carbon and 0.02g of hydrogen. (RAMs of C =12; H = 1)

(3 marks)

i. Explain the following terms:

1. electrolysis

_____

_____

(1 mark)

**Name of a Candidate: _____**

2. electroplating

_____

_____

(1 mark)

ii. What is the chemical formula for rust?

_____

(1 mark)

7. (a) Study the condensed structures of some organic compounds **D**, **E**, **F** and **G** given below and use them to answer the questions that follow:

**D.** $CH_3(CH_2)_4CH_3$ **E.** HCOOH **F.** $CH_3CH_2CH_2CH_2OH$ **G.** $CH_3CH = CH_2$

i. Name the family of organic compounds to which each of the compounds above belongs.

D. _____ E. _____ F. _____ G. _____

(4 marks)

ii. Describe a chemical test you can carry out to distinguish compound **E** from compound **F**.

_____

_____

_____

(3 marks)

(b) i. Define 'esterification'

_____

_____

(1 mark)

ii. Write down the chemical equation of the reaction between methanoic acid and ethanol.

_____

_____

(2 marks)

iii. Give the name of the main product in the reaction above.

_____

(1 mark)

iv. State any **two** uses of esters.

_____

_____

(2 marks)

**Name of a Candidate: _____**

8. (a) i. What is radioactive decay?

_____

_____

(1 mark)

ii. Mention **two** differences between an alpha particle and a gamma ray?

_____

_____

_____

_____

(2 marks)

iii. State **one** use of nuclear radiation in agriculture

_____

(1 mark)

(b) Lead (Pb) – 120 (atomic number 82) isotope undergoes radioactive decay to form Bismuth (Bi) – 120 (atomic number 83).

i. What type of radioactive decay did Lead (Pb) undergo?

_____

(1 mark)

ii. Write the nuclear equation to show the change stated above in (b).i.

_____

_____

(3 marks)

(c) A radio-isotope **Y** has a half-life of 25 minutes. A sample of **Y** initially contains 600g.

i. What does the term "half-life" mean?

_____

_____

(1 mark)

ii. How much mass of Y will remain after 100 minutes?

(4 marks)

***END OF QUESTION PAPER***

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EN DOF QUESTION PAPER